Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	A technology based on frugal innovation principles that allows in-situ digital sensor integration in biomanufacturing processes.			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Electrical・Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Bioprocesses / Bioequipment	Bioprocess Equipment Technologies	Electrical and Electronic Components	Sensor Components
Concept of Proposed Technology	We propose a wireless sensing technology based on frugal innovation principles that leverages research and development of smart components for bioprocess equipment. Frugal innovations [1] are customized products or services that are functional, robust, resource-efficient, sustainable, affordable, easy to use, low-maintenance, and yet technically at a high level although the design can look very simply. The core concept evolves around battery-free resonant circuits, embedded in the housing material of bioprocess equipment, preferably (single-use) components such as filters or different kinds of bags. The proposed technology allows for wireless sensing of environmental factors such as conductivity, pressure and temperature. Further targeted parameters can be pH and biomass for metabolic monitoring. The resonant frequency of the small sensor component encodes the sensor data. Miniaturized readout electronics enable IoT connectivity with digital plants, ensuring compatibility with digital process control systems.			

	This technology ensures low-cost scalability, predictive maintenance, and complete traceability within biomanufacturing processes. Advanced in-situ monitoring helps to objectively record and control the bioprocess in real time, which is crucial for performance, optimization, and efficiency.
Rationale for Joint R&D Project	The problem we are facing is that biomanufacturing processes such as downstream purification still rely on single-use consumables such as filters or different kinds of single-use bags that are disconnected from digital process control, which is a critical barrier to full digital integration. Although there are state-of-the-art in-line sensors and sensor monitors for pressure, temperature, conductivity and other parameters in place, they have some drawbacks, which limit their use to create smart components for bioprocess equipment. State-of-the-art sensors and monitors have bulky dimensions, have a lot of cable connections, need time-consuming installation and configuration, need to fit different tubing sizes, and are very expensive. Collaboration with SME partners is required: • for the development and manufacture of the sensor transducer or sensing materials to produce capacitive or resistive changes in the battery-free resonant circuits. • to adapt technology to specific applications in Korea, support application-specific connectivity to a bioprocess control system, provide local manufacturing capabilities, regulatory knowledge, as well as market access, and lead commercialization in Korea. • to transform the proposed technology into a marketable product and establish market entry. The proposed joint R&D project enables knowledge exchange, shortens time to market, and creates a commercially viable platform through cross-border collaboration.
Key Words	1. Frugal innovation 2. Wireless, battery-free sensing 3. Digital transformation 4. Complete wireless connectivity (no additional probes) 5. Edge intelligence

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (5)
R&D Objectives	We aim to develop a working prototype for a smart component for bioprocess equipment such as a smart downstream filter based on the principle of frugal innovation. Such component splits into the following parts that will be developed through this R&D project: • **Wireless sensor**: The frugal innovation principle uses minimal electronic parts (no battery, no IC chips, no programming) to create a low-cost passive resonant circuit for wireless sensing. Such circuits are based on transducers or materials that are well-suited to design structures which are sensitive to environmental parameters such as pressure, conductivity and temperature. The sensor circuit is inductively coupled to the customized readout module. The resonator coil is used as the sensor antenna. Sensor data is encoded in the resonant frequency. • **Readout module**: System intelligence is shifted to the readout electronics module. It consists of a microcontroller and embedded software for signal processing and wireless communication interfaces such as WiFi. Essential is an algorithm for rapid sensor signal processing in time domain that simplifies electronic implementation. • **Connectivity**: The miniaturized readout module has IoT connectivity to the digital process control system resulting in enhanced process reliability, because of the seamless digital integration of the proposed smart bioprocess equipment. • **Integration**: The wireless sensor and readout module will be integrated into a prototype for a smart component for bioprocess equipment.			

R&D Target Specifications	• The small low-cost wireless sensor can be embedded in the plastic housing material of bioprocess equipment. The estimated material cost for prototyping is about 2 € and can get cheaper for mass production. Its exemplary diameter can be about 16 mm (size of a 1 cent coin).
	• The readout electronics module can sit on the smart component housing; the size is compact with exemplary 9 x 4 cm and the estimated material cost for prototyping is about 30 €.
	• Target sensor specification:
	  ○ Conductivity: ±0.25 mS, 1 µS/cm to 200 mS/cm range
	  ○ Pressure: ±0.3 psi, 0-60 psi range
	  ○ Temperature: ±0.5 °C, 4 °C to 70 °C range			

R&D Plan	Research approach:
	• Total duration: 36 months (3 years).
	• Phase I (Months 1–6): Create a clear roadmap for the project
	  ○ Define project scope and objectives
	  ○ Establish roles and responsibilities
	  ○ Prepare system requirements and system design (WP1)
	• Phase II (Months 7–36): Collaborative research and development (WP2-WP6)
	  ○ Leverage combined strengths for innovative solutions
	  ○ Jointly develop the stated R&D objectives
	  ○ Utilize Fraunhofer's expertise in application-specific research and system integration
	  ○ Conduct tests and validate functionality
	  ○ Gather feedback from end-users in Korea			

	○ Adjust to local manufacturing and regulatory standards ○ Prepare for future market entry Work packages: • WP1 - System requirements and system design (Month 1-6) • WP2 - Development of low-cost wireless sensors hardware (Month 7-18) • WP3 - Development of miniaturized readout electronics hardware (Month 10-21) • WP4 - Development of signal processing and programming of embedded software (Month 16-27) • WP5 - System integration, prototype setup, testing, and documentation (Month 26-36) • WP6 - Technology transfer and dissemination planning (Month 31-36) Milestones and Timeline: • MS1 - System specifications and prototype concept are available (Month 6 – End of Phase I) • MS2 - Assembled wireless sensor hardware is available and verified (Month 18) • MS3 - Assembled readout electronics hardware is available and verified (Month 21 – Half of Phase II) • MS4 - Embedded programming of signal processing and communication interfaces completed (Month 27) • MS5 - System integration and prototype evaluation completed (Month 36 – End of Phase II)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Provide expertise in application-oriented research, resonant circuits, readout electronics and algorithm development as well as system integration. • Lead technical research and innovation aspects. • Provide extensive experience in the development of miniaturized electronics for harsh environments. • Evaluate the scalability of the technology. • Use of professional electronics engineering tools for simulation and PCB design. • Develop prototypes for in-situ sensor integration. • Provide laboratories with professional electronics measurement equipment. • Conduct rigorous testing and validation of the technology. • Provide expertise on compliance with industry standards. • Support in assessing the technology's market potential. • Help in developing a commercialization strategy. • Provide electrical engineering and bioengineering personnel as well as project management support.
Expected R&R of Korean SME Partner	• Contribute to practical applications and user needs. • Provide insights into market trends and requirements. • Tailor the technology for specific bioprocess equipment or other relevant applications. • Assist in the design and manufacturing of prototypes. • Provide local manufacturing capabilities. • Facilitate real-world testing in biomanufacturing settings. • Assist in navigating local regulatory landscapes. • Conduct market research and identify target customers. • Promote early adoption among potential customers. • Convert the proposed technology into a market-ready product and secure market entry.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• A very cost-effective technology that creates a new class of sensors that bridges the gap between conventional electronics and disposable material systems. • A technology based on frugal innovation principles, which gives new design and sensor integration options for smart bioprocess equipment. • Since the sensor is completely passive and sterilizable, it is ideal for use in disposable bioprocess equipment where conventional electronics cannot be used. • Support the increasing digital transformation in biomanufacturing processes and enable cost-efficient inline monitoring without interfering with sterile environments. • An evaluated prototype tailored to the Korean market. • Strengthened collaboration and knowledge exchange between Fraunhofer and the Korean SME. • A sustainable model for future joint projects.

6. References

(1) N. Agarwal, J. Oehler and A. Brem, "Constraint-Based Thinking: A Structured Approach for Developing Frugal Innovations," in IEEE Transactions on Engineering Management, vol. 68, no. 3, pp. 739-751, June 2021, doi: 10.1109/TEM.2020.3042929.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization